Project Report: Global Layoffs Data Analysis (2020–2023)

1. Project Overview

- **Goal:** To analyze the global layoff landscape from 2020 to 2023, identifying trends across different industries, locations, and company stages.
- **Primary Focus:** Identifying the core reasons behind mass layoffs and the correlation between company funding and employee terminations.

2. Dataset Summary

- **Source:** A comprehensive dataset of global layoffs. With 1932 rows and 9 columns.
- **Key Features:** Company Name, Industry, Location, Total Laid Off, Percentage, Date, Stage, and Funds Raised(Millions).

3. Data Cleaning and Transformation using SQL

- **Handling Duplicates:** Removing identical records from the layoffs_staging3

```sql
-- DATA CLEANING
CREATE TABLE `layoffs_staging3` (
  `company` text,
  `location` text,
  `industry` text,
  `total_laid_off` double DEFAULT NULL,
  `percentage_laid_off` double DEFAULT NULL,
  `date` text,
  `stage` text,
  `country` text,
  `funds_raised_millions` double DEFAULT NULL,
  `row_num` INT                                    -- create a new column to find the duplicate values
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```sql
INSERT INTO layoffs_staging3
SELECT *,
ROW_NUMBER() OVER(PARTITION BY company, location, industry, total_laid_off,
percentage_laid_off, `date`, stage, country, funds_raised_millions) AS row_num
FROM layoffs_staging;
-- we can find all duplicates values to notice the column `row_num` where >1 is duplicate then delete this value
DELETE
FROM layoffs_staging3
WHERE row_num > 1;
```

- **Standardization:** Fixing industry names (e.g., merging 'Crypto Currency' , ‘CryptoCurrency' into 'Crypto') and trimming whitespace.

```sql
SELECT * FROM layoffs_staging3
WHERE industry LIKE 'Crypto%';
```

```sql
UPDATE layoffs_staging3
SET industry = 'Crypto'
WHERE industry LIKE 'Crypto%';
```

- **Null Value Treatment:** Populating missing industry data based on existing company records and handling nulls in `total_laid_off`.

  - ```
    SELECT * FROM layoffs_staging3
    WHERE total_laid_off IS NULL AND
    percentage_laid_off IS NULL;
    ```

  - ```
    DELETE FROM layoffs_staging3
    WHERE total_laid_off IS NULL AND
    percentage_laid_off IS NULL;
    ```

- **Feature Engineering:** Converting text-based dates into standard `DATE` format for time-series analysis.

  - ```
    SELECT `date`,
    str_to_date(`date`,'%m/%d/%Y')
    FROM layoffs_staging3;
    ```

  - ```
    UPDATE layoffs_staging3
    SET `date` = str_to_date(`date`,'%m/%d/%Y');
    ```

  - ```
    SELECT * FROM layoffs_staging3;
    ```

  - ```
    ALTER TABLE layoffs_staging3
    MODIFY COLUMN `date` DATE;
    ```

4. Advanced SQL Analysis (Business Insights)

- **The Giants:** List of top 5 companies with the largest layoffs in a single event.

company	total_laid_off	percentage_laid_off	date
Google	12000	0.06	2023-01-20
Meta	11000	0.13	2022-11-09
Microsoft	10000	0.05	2023-01-18
Amazon	10000	0.03	2022-11-16
Ericsson	8500	0.08	2023-02-24

- **Yearly Impact:** Total layoffs broken down by 2020, 2021, 2022, and 2023.

year	layoffs
2020	80416
2021	15823
2022	159517
2023	123085

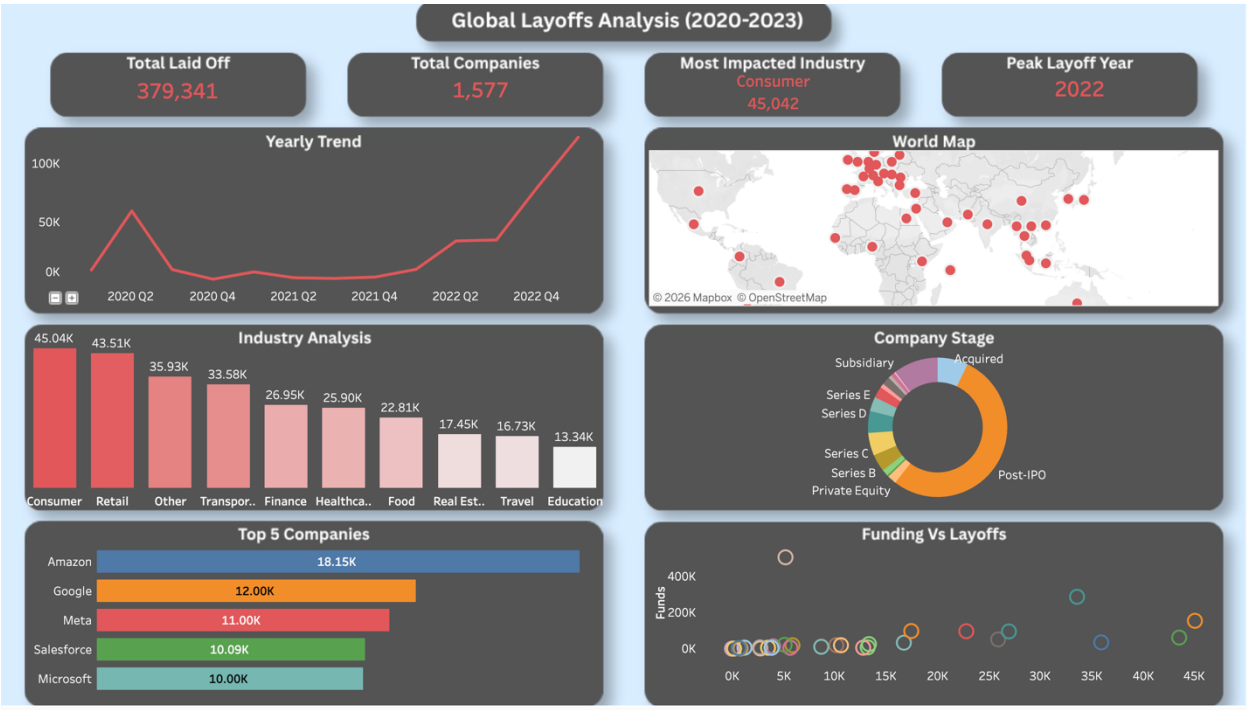
- **Company Stages:** Analysis of which stage (Post-IPO, Series B, etc.) is most vulnerable.

stage	layoffs
Post-IPO	202266
Unknown	38907
Acquired	27130
Series C	20017
Series D	19225
Series B	15311
Series E	12697
Series F	9915
Private Equity	7957
Series H	7244
Series A	5678
Series G	3697
Series J	3570
Series I	2855
Seed	1536
Subsidiary	1014

- **Rolling Totals:** A cumulative monthly look at layoffs to see the trend's growth.

month	layoffs	rolling_to...
2020-03	9298	9298
2020-04	26557	35855
2020-05	25783	61638
2020-06	7627	69265
2020-07	7112	76377
2020-08	1969	78346
2020-09	609	78955
2020-10	450	79405
2020-11	237	79642
2020-12	774	80416
2021-01	6813	87229
2021-02	868	88097
2021-03	47	88144
2021-04	261	88405
2021-06	2434	90839
2021-07	80	90919
2021-08	1867	92786
2021-09	161	92947
2021-10	22	92969
2021-11	2070	95039
2021-12	1200	96239
2022-01	510	96749
2022-02	3685	100434
2022-03	5714	106148
2022-04	4078	110226
2022-05	12885	123111
2022-06	17294	140405
2022-07	16171	156576
2022-08	13055	169631
2022-09	5881	175512
2022-10	17406	192918
2022-11	52662	245580
2022-12	10176	255756
2023-01	82349	338105
2023-02	36316	374421
2023-03	4420	378841

5. Interactive Visualizations (Tableau Dashboard)



- **KPI Overview:** Displaying Total Laid Off, Total Companies, and Peak Year.
- **Industry Trends:** Bar charts showing the hardest-hit sectors like Retail and Consumer.
- **Funding vs. Layoffs:** Scatter plots demonstrating the relationship between funds raised and layoffs.

6. Conclusion & Recommendations

6.1. Project Summary

This analysis provided a clear picture of the global layoff crisis from 2020 to 2023. By cleaning data in MySQL and visualizing it in Tableau, we identified which industries and company stages were hit the hardest.

6.2. Key Findings

- **Peak Crisis Period:** Data shows that layoffs reached their absolute peak in late 2022 and early 2023, far surpassing the initial 2020 pandemic wave.

- **Most Impacted Industries:** Retail and Consumer sectors faced the highest volume of job cuts compared to other industries.

- **The Funding Myth:** Having large amounts of funding did not protect employees; many highly-funded companies still executed mass layoffs to manage their burn rates.

- **Company Vulnerability:** Large "Post-IPO" companies were responsible for the biggest layoff events, while smaller startups often faced total shutdowns.

6.3. Strategic Recommendations

- **Sustainable Growth:** Companies should focus on stable, long-term growth instead of aggressive hiring fueled by temporary funding.

- **Skill Diversification:** Employees in high-risk sectors (like Tech and Retail) should be encouraged to learn diverse skills to remain employable during market shifts.

- **Proactive Monitoring:** Businesses should use data-driven dashboards (like the one built in this project) to monitor market trends and prepare for economic downturns before they happen.

- **Financial Safety Nets:** Investors should encourage companies to maintain better cash reserves to protect their workforce during "Peak Layoff" years.